

Worksheet 1- Solutions

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1. It is asked for the estimation of the lamps' defection proportion. So that the difference between the exact and estimated proportion is not more than 0.03, determine the sample size for the 95% confidence interval

- a) If we know from past experiences that the sample proportion is 0.05.
b) If we have no idea about sample proportion.

$$a) P[\hat{p} - e < \hat{p} < \hat{p} + e] = 1 - \alpha, \quad e = 0.03, \quad \hat{p} = 0.05, \quad \hat{q} = 0.95, \quad z_{0.025} = 1.96$$

$$e = z_{\alpha/2} \sqrt{\frac{pq}{n}} \Rightarrow n = \left\lceil \frac{z_{\alpha/2}^2 pq}{e^2} \right\rceil$$

$$b) \quad n = \left\lceil \frac{1.96^2 \times 0.05 \times 0.95}{0.03^2} \right\rceil = \left\lceil 202.751 \right\rceil = 203$$

$$\max \left(\frac{z_{\alpha/2}^2}{e^2} p(1-p) \right) \Rightarrow \frac{d}{dp} \left(\frac{z_{\alpha/2}^2}{e^2} (p-p^2) \right) = \frac{z_{\alpha/2}^2}{e^2} (1-2p) = 0$$

$$n = \left\lceil \frac{1.96^2 \times 0.5 \times 0.5}{0.03^2} \right\rceil = \left\lceil 1067.11 \right\rceil = 1068 \Rightarrow p = 1/2$$

2. The diameter of the screws produced in a factory has a normal distribution with a standard deviation of 0.38 cm. To estimate the mean of screw diameters, 12 random screws are selected, and the 95% confidence interval is computed as

$$1.64 \leq \mu \leq 2.36$$

However, the factory manager asks for a more definite interval. So that the difference between the exact value and the estimated value is not more than 0.08 cm, determine the sample size for

- a) 95% confidence interval.
b) 99% confidence interval.

$$a) \quad 1.64 \leq \mu \leq 2.36 \Rightarrow 2e = 2.36 - 1.64 \Rightarrow e = 0.36 \Rightarrow \left[2 \pm 0.36 \right] \Rightarrow e = 0.08$$

$$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$e = z_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$\Rightarrow n = \left\lceil \frac{z_{\alpha/2}^2 s^2}{e^2} \right\rceil = \left\lceil \frac{1.96^2 \times 0.38^2}{0.08^2} \right\rceil = \left\lceil 86.68 \right\rceil = 87$$

$$b) \quad \alpha = 0.01$$

$$n = \left\lceil \frac{2.58^2 \times 0.38^2}{0.08^2} \right\rceil = \left\lceil 150.185 \right\rceil = 151$$

3. X_1 and X_2 represent the elements of two samples that are randomly selected from a population whose mean is μ and variance is σ^2 . Four estimators of μ are given below:

$$\hat{\mu}_1 = \frac{1}{2}X_1 + \frac{1}{2}X_2, \quad \hat{\mu}_2 = \frac{2}{5}X_1 + \frac{3}{5}X_2, \quad \hat{\mu}_3 = \frac{1}{5}X_1 + \frac{2}{5}X_2, \quad \hat{\mu}_4 = \frac{1}{4}X_1 + \frac{3}{4}X_2$$

- a) Find the unbiased estimators.
b) Which estimator is the most effective?
c) Compute the relative efficiency of $\hat{\mu}_1$ with respect to $\hat{\mu}_2$ and interpret the results.

$$a) \quad E(\hat{\mu}_1) = E\left(\frac{1}{2}X_1 + \frac{1}{2}X_2\right) = \frac{1}{2}\mu + \frac{1}{2}\mu = \mu \quad \checkmark$$

$$E(\hat{\mu}_2) = \frac{2}{5}\mu + \frac{3}{5}\mu = \mu \quad \checkmark$$

$$E(\hat{\mu}_3) = \frac{1}{5}\mu + \frac{2}{5}\mu = \frac{3}{5}\mu \quad \times$$

$$E(\hat{M}_3) = \frac{1}{5}M + \frac{2}{5}M = \frac{3}{5}M \times$$

$$E(\hat{M}_4) = \frac{1}{4}M + \frac{3}{4}M = M \checkmark$$

$$b) \left. \begin{aligned} \text{Var}(\hat{M}_1) &= \text{Var}\left(\frac{1}{2}x_1 + \frac{1}{2}x_2\right) = \frac{1}{4}\sigma^2 + \frac{1}{4}\sigma^2 = \frac{1}{2}\sigma^2 \\ \text{Var}(\hat{M}_2) &= \frac{4}{25}\sigma^2 + \frac{9}{25}\sigma^2 = \frac{13}{25}\sigma^2 \\ \text{Var}(\hat{M}_4) &= \frac{1}{16}\sigma^2 + \frac{9}{16}\sigma^2 = \frac{10}{16}\sigma^2 \end{aligned} \right\} \hat{M}_1 \text{ is the most efficient.}$$

$$c) \text{Rel. Eff.} = \frac{\text{Var}(\hat{M}_2)}{\text{Var}(\hat{M}_1)} = \frac{\frac{13}{25}\sigma^2}{\frac{1}{2}\sigma^2} = \frac{26}{25} = 1.04$$

If we use $\hat{M}_2$ instead of $\hat{M}_1$, we need a sample of size $1.04n$ where n is the sample size in case that $\hat{M}_1$ is used.

4. An operator's long-distance phone calls are normally distributed with a mean of 8 minutes and a standard deviation of 2 minutes. If you choose samples of 25 calls, what percentage of these sample means will be between 7.8 and 8.2 minutes?

$$X \sim N(8, 2^2), \quad n = 25$$

$$P(7.8 < \bar{x} < 8.2) = P\left(\frac{7.8-8}{2/\sqrt{25}} < z < \frac{8.2-8}{2/\sqrt{25}}\right) = P(-0.5 < z < 0.5)$$

$$= 2 \times 0.1915 = 0.383$$

$$\Rightarrow 38.3\%$$

5. An n -unit sample will be taken from a population that is normally distributed with a standard deviation of 8. What n should be so that the difference between the sample mean and the population mean is at most 1.5 with a probability of 0.95?

$$X \sim N(\mu, 8^2), \quad P\left[\bar{x} \pm \underbrace{z_{0.025} \frac{8}{\sqrt{n}}}_{c=1.5}\right] = 0.95$$

$$n = \left\lceil \frac{1.96^2 \times 8^2}{1.5^2} \right\rceil = \left\lceil 109.27 \right\rceil = 110 //$$

6. A sample of size 400 units is selected without replacement from a population of size 5000 units. The standard deviation of the population is $\sigma = 50$. The sample mean is calculated as 4500. Estimate the population mean with 0.90 confidence level.

$$\frac{n}{N} = \frac{400}{5000} = 0.08 > 0.05, \quad \sigma = 50, \quad \bar{x} = 4500, \quad n \geq 30$$

$$\bar{x} \pm z_{0.05} \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \Rightarrow \left[4500 \pm 1.645 \frac{50}{20} \sqrt{\frac{4600}{4999}} \right]$$

$$= [4496.66, 4503.44] //$$

7. The weights of 10000 products in a warehouse are examined. A sample of 36 units of these products was selected (without replacement), the mean of which was 20 gr. and its standard deviation is 4 gr. found. Estimate the weight mean of the products in the warehouse with confidence level 0.70.

$$\begin{aligned}
 N &= 10000, & n &= 36, & \bar{X} &= 20, & S &= 4, & \bar{X} &\pm z_{0.15} \frac{S}{\sqrt{n}} \\
 \frac{n}{N} &= \frac{36}{10000} < 0.05, & & & & & & & & [20 \pm 1.04 \frac{4}{6}] = [19.31, 20.69]
 \end{aligned}$$

8. If $N=5000$, $n=49$, $\sum_{i=1}^{49} X_i = 2940$ and $\alpha=0.05$, then determine confidence interval for μ . ($\sigma = 7$)

$$\begin{aligned}
 \left[\bar{X} \pm z_{\alpha/2} \frac{S}{\sqrt{n}} \right] &= \left[\frac{2940}{49} \pm 1.96 \frac{7}{7} \right] = [58.04, 61.96] \\
 \downarrow & \\
 \frac{\sum X_i}{n} &
 \end{aligned}$$

9. The proportion of defective products produced by machine A is investigated. For this purpose, 180 out of 2000 products produced by the machine were chosen randomly, and it was understood that 27 of them were faulty. Estimate the proportion of defective products produced by machine A with a 99% confidence interval (selection is made without replacement).

$$\begin{aligned}
 n &= 180, & N &= 2000, & \hat{p} &= \frac{27}{180}, & \frac{n}{N} &= \frac{180}{2000} = 0.09 > 0.05 \\
 & & & & & & & & & \left[\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}} \sqrt{\frac{N-n}{N-1}} \right] \\
 & & & & & & & & & \left[\frac{27}{180} \pm 2.575 \sqrt{\frac{\frac{27}{180} \times (1 - \frac{27}{180})}{180}} \sqrt{\frac{2000 - 180}{2000 - 1}} \right] = [0.085, 0.215]
 \end{aligned}$$